

Generation Technology Cost Assumptions

Senegal PyPSA-Earth model: benchmarking of solar, wind, hydro, and thermal generation inputs against real-world sources, and correction of fuel costs

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1. Overview

The PyPSA-Earth pipeline sources generation technology costs from the `technology-data` project (v0.13.2), which in turn compiles figures from a mix of sources per technology — predominantly the Danish Energy Agency (DEA) *Technology Data Catalogue for Generation of Electricity and District Heating*, with some older carriers (CCGT, OCGT, oil, coal, biomass fuel costs) backfilled from a 2013 DIW working paper itself based on 2010–2011-vintage IEA data.

This document records the benchmarking exercise carried out against independent external sources for every non-storage generation carrier in the model (`solar`, `onwind`, `hydro`, `CCGT`, `OCGT`, `oil`, `coal`, `biomass`) across six parameters — **investment (CapEx)**, **lifetime**, **efficiency**, **fixed O&M (FOM)**, **variable O&M (VOM)**, and **fuel cost** — and what, if anything, was corrected as a result.

Only fuel cost was corrected in this pass (`gas`, `coal`, `biomass`; `oil` was checked and left unchanged). Investment, lifetime, efficiency, and FOM/VOM were all benchmarked and are reported below, but were **not** changed: in most cases the model's figures were found to be reasonably close to

at least one credible reference, and where a real gap was found (onshore wind CapEx and FOM, biomass efficiency and VOM), it is recorded here as a flagged, open item for a future sensitivity run rather than corrected outright, for the same reason described in the companion BESS document — every changed number in this project traces to a specific source, and no clean, current, Senegal-specific benchmark could be found for those particular figures to correct *to*.

Battery storage (BESS) inputs are **not** re-covered here in depth — that recalibration (CapEx, FOM, VOM/degradation, efficiency) was done separately and is documented in *BESS_cost_assumptions* (Section 8 below gives a one-paragraph summary for completeness).

All fuel cost changes are applied via `costs.fuel:` overrides in each scenario's config file (`config.uc_bess_{year}.yaml` / `config.uc_no_bess_{year}.yaml`, years 2025–2050), flowing through `process_cost_data.py`'s override loop into the resolved cost table used by `add_electricity.py`.

2. Investment (CapEx) — checked, not changed

2.1 Method

Model figures (from `technology-data`, transcribed with full per-parameter sourcing in `input_cost_data.xlsx`) were compared against four independent tiers, all expressed on an overnight-CAPEX (USD/kW) basis at a flat EUR→USD rate of 1.08 (nominal comparison only, not inflation-adjusted to each source's vintage year):

1. **Global benchmark** — IRENA *Renewable Power Generation Costs 2024* (solar/wind), Lazard *LCOE+ v18.0* (2025) (CCGT).
2. **EIA AEO2022**, Table 8.2 — a second independent global anchor, standardized across nearly every conventional and renewable technology in one document (2021 USD/kW, Sargent & Lundy basis).
3. **Senegal-calibrated benchmark** — UTS *Africa Power Report: Senegal* (2025).
4. **Senegal actual-project benchmark** — Taiba N'Diaye wind farm (the one real Senegalese transaction available).

2.2 Findings

technology	model (USD/kW)	global	EIA	Senegal (UTS/actual)
solar	441	691 (IRENA)	1,327	989
onwind	1,101	1,041 (IRENA)	1,718	2,085 (Taiba N'Diaye, actual)
hydro	2,457	—	3,083	2,674
CCGT	914	2,500 (Lazard)	1,062	504

technology	model (USD/kW)	global	EIA	Senegal (UTS/actual)
OCGT	471	—	785	—
oil	384	—	2,018	908
coal	4,133	—	4,074	2,018
biomass	2,525	—	4,524	2,371

- **Onshore wind is the most consequential finding in this section.** Three independent “generic reference” sources cluster near \$1,000–1,700/kW (model \$1,101, IRENA \$1,041, EIA \$1,718), while the one real Senegalese transaction — Taiba N’Diaye — sits at \$2,085/kW, well above all three. This is the strongest evidence anywhere in the benchmarking exercise that a first-of-kind African wind project costs meaningfully more than any generic reference figure, Senegal-calibrated or not. **Flagged, not corrected:** no second Senegalese wind transaction exists to confirm Taiba N’Diaye is representative rather than a one-off, so raising the model’s figure to match a single data point was judged premature.
- **CCGT:** EIA (\$1,062/kW) sits close to the model (\$914/kW); Lazard’s recent US quotes (\$2,500/kW) are the outlier — most plausibly recent US gas-turbine supply-chain inflation that does not generalise to Senegal. Model figure judged reasonable.
- **Coal:** EIA (\$4,074/kW) and the model’s Lazard-sourced figure (\$4,133/kW) agree closely. Senegal has no operating utility-scale coal plant, so there is no real transaction to adjudicate between this pair and the lower UTS figure (\$2,018/kW).
- **Biomass:** model figure (\$2,525/kW) sits close to the Senegal-calibrated UTS figure (\$2,371/kW); EIA’s figure (\$4,524/kW) is nearly 80% higher — a reminder that “matches the Senegal reference” and “matches the global reference” can point in different directions, and the Senegal-specific comparison is the more decision-relevant one here.
- **Source vintage does not cleanly predict deviation.** Plotting each technology’s `currency_year` against its deviation from the global benchmark shows no clean relationship — a 2011-vintage transmission figure and a 2020-vintage storage figure both sit far from current benchmarks, but technology-specific learning-curve speed confounds vintage: fast-learning technologies (batteries, and to a lesser extent solar) need re-benchmarking far more often than slow-learning ones (hydro, biomass, oil) from the same source year.

No CapEx values changed. Onshore wind is the one open item worth a sensitivity run before the thesis is finalised, since wind is actually built in both solved networks (unlike coal and biomass, discussed below).

3. Lifetime — checked, not changed

Compared against a reference-country capacity-expansion case (Indonesia — MEMR 2024 / Langer et al. 2024, via one TU Delft thesis; Athanasiou catalogue, via another). Not Senegal-specific — a

plausibility check, not validation, chosen because it is the only source found reporting lifetime consistently for nearly the whole technology set.

technology	model (years)	reference (years)
solar	40	30
onwind	30	30
hydro	80	50
CCGT	25	25
OCGT	25	25
oil	25	25
coal	40	30
biomass	30	25

Four exact matches (CCGT, OCGT, oil at 25 years; onshore wind at 30) suggest the thermal and wind lifetime assumptions are conventional, industry-standard figures. The two gaps both run the same direction — model longer than reference — for hydro (80 vs. 50y) and coal (40 vs. 30y); both are old-vintage DIW/Lazard entries using a more optimistic technical life than the 2024 reference study assumes. **Not changed** — worth a footnote if either materially affects a levelised-cost result, but neither carrier’s lifetime was judged consequential enough on its own to correct without a clearer Senegal-specific anchor.

4. Efficiency — checked, not changed (biomass flagged)

Thermal and hydro efficiency only (solar/wind/battery/transmission excluded — “efficiency” is set to 1 by convention for those and is not a meaningful cross-source metric). Reference-case figures from the same Indonesia MEMR 2024 source as Section 3; a second, fully independent check via EIA AEO2022 heat rates (converted at 3,412 Btu = 1 kWh — note this may not use exactly the same accounting basis, HHV vs. LHV, as the DEA figures underlying the model).

technology	model	reference	EIA (heat-rate implied)
CCGT	60.0%	61.0%	53.6%
OCGT	43.0%	40.0%	34.4%
oil	35.0%	48.0%	41.1%
coal	35.6%	37.0%	39.5%
biomass	46.8%	32.0%	25.3%
hydro	90.0%	95.0%	—

Biomass is the clear anomaly. The model’s 46.8% is 10–15 percentage points above *both* independent benchmarks, and reads closer to a modern combined-cycle gas efficiency than a solid-biomass steam cycle (literature consistently reports 25–35% for that technology class). Everything else is close enough to be unremarkable — CCGT and coal both track their reference figures within a few points, and the OCGT/hydro gaps run in plausible opposite directions.

Not changed. Biomass has never been built in any solved network in this project to date, so this doesn’t affect current results, but it is flagged as the figure most worth checking against the underlying DIW source directly (possible transcription issue upstream in `technology-data`) before being cited or relied on for anything beyond illustration.

5. Fixed O&M (FOM) — checked, not changed

Model FOM is `%/year of investment`. External sources report `$/kW/year`, converted to a percentage using *that source’s own* investment figure (not the model’s), so the arithmetic stays auditable rather than mixing bases.

technology	model	reference	EIA
solar	2.1%	1.0%	1.2%
onwind	1.2%	3.0%	1.6%
hydro	1.0%	1.9%	1.4%
CCGT	3.2%	2.5%	1.2%
OCGT	1.8%	2.4%	0.9%
oil	2.4%	1.0%	1.8%
coal	1.3%	2.7%	1.0%
biomass	4.5%	2.4%	2.9%

No systematic direction: the model sits above both references for CCGT and biomass, below both for onshore wind and coal, and in between for the rest — consistent with each `technology-data` entry being sourced independently rather than adjusted together as a coherent set. **Onshore wind stands out most:** the model’s 1.2%/year is roughly a third of the reference case’s 3.0%/year, pairing with the CapEx finding in Section 2 — both point the same direction, that the model may understate the full cost of operating wind in a market like Senegal’s, not just the cost of building it.

Not changed, for the same reason as CapEx: flagged as a sensitivity-run candidate rather than corrected against a non-Senegal reference case.

6. Variable O&M (VOM) — checked, not changed (biomass flagged)

Directly comparable in \$/MWh (excluding fuel) — no unit conversion needed.

technology	model	reference	EIA
solar	0.0	0.0	0.0
onwind	1.3	0.0	0.0
hydro	0.0	0.7	1.5
CCGT	4.2	2.5	2.0
OCGT	4.8	3.5	4.7
oil	6.3	6.8	6.0
coal	3.3	1.2	4.7
biomass	0.0	3.1	5.1

Oil/diesel is the strongest match found anywhere in this benchmarking exercise — model 6.35, reference case 6.84, both diesel reciprocating plants, independently sourced, within 8% of each other. **Biomass’s zero VOM is the standout gap:** both references carry a real, nonzero variable cost (3–5/MWh), and a true zero is unusual for any combustion technology with ash handling and consumables. Combined with the efficiency anomaly in Section 4, biomass is the technology most worth a closer look at its underlying DIW source before it is used for anything beyond illustration.

Not changed — biomass currently plays no role in either solved network, so this is a “check before you lean on it” flag, not an urgent fix.

7. Fuel cost — checked and corrected

7.1 Why this parameter, specifically

Fuel cost feeds directly into `marginal_cost` (= VOM + fuel/efficiency) and had not been checked anywhere else in this benchmarking pass, despite Senegal’s generation fleet still being overwhelmingly dependent on imported liquid fuel (oil- and gas-fired plants supplied roughly 80% of generation in 2024). A stale fuel price assumption has more leverage over model outcomes than almost any other single parameter, since it directly shapes how often thermal generation is dispatched instead of BESS in every solved network in this project.

7.2 Model defaults (`technology-data` , pre-correction)

All four fuel entries are static catalogue values with no year-projection trajectory attached — unlike CapEx, which has a full year-by-year table, `fuel` is carried forward flat regardless of the selected cost-year:

carrier	model default (EUR/MWh_th)	source	vintage
gas (→ CCGT, OCGT)	24.57	DIW (2013), based on IEA 2011 data	2010
coal	9.55	DIW (2013), based on IEA 2011 data (~99 USD/t)	2010
biomass	7.41	IEA (2011b), old PyPSA cost assumptions	2015
oil	52.91	IEA WEM2017, 97 USD/boe projection	2015

`costs.at["OCGT","fuel"] = costs.at["gas","fuel"]` and the equivalent for CCGT run in `process_cost_data.py` after any override, so correcting `gas` alone is sufficient to correct both gas-fired carriers automatically.

7.3 Benchmarks checked

- **Oil:** Brent crude (~\$79.7/bbl, Aug 2026) and Rotterdam/Singapore/Fujairah HSFO 380 bunker fuel (~\$550/mt, the closer proxy to what a Senegalese HFO plant actually burns), both converted to EUR/MWh_th. Senegal’s own “Gas to Power” strategy note (Ministry of Petroleum and Energy) independently cites \$14/MMBtu HFO as its own fossil comparator.
- **Gas:** initially checked against three international hub benchmarks — Henry Hub (US domestic), TTF (European), and JKM (Asian LNG spot) — but these were judged the wrong comparison: Senegal’s actual future gas supply is **domestic and piped** (GTA, Sangomar, Yakaar/Teranga), not internationally-traded LNG, and the hub benchmarks are additionally distorted by an active Gulf LNG supply disruption at time of writing. Used instead: Senegal’s own “Gas to Power” strategy note, which cites **\$5–6/MMBtu domestic gas**.
- **Coal:** API2 Rotterdam benchmark (~\$120/t, Aug 2026, 6,000 kcal/kg NAR = 25.1 GJ/t). No landed Senegal-specific coal price found — Sendou’s coal is imported — so this is used as the best available proxy without a further (unquantified) shipping/import premium.
- **Biomass:** Lamers et al. (2015) agricultural-residue feedstock cost range, already cited in the project’s own UTS *Africa Power Report: Senegal* — deliberately **not** the EU wood-pellet market benchmark (37–64 EUR/MWh_th), since Senegal’s dominant available feedstock (groundnut shell, 700,000–900,000 t/year in the groundnut basin) is a near-waste agricultural residue, not commercially-traded wood pellets — a different market entirely.

7.4 Corrections applied

carrier	old default (EUR/MWh_th)	new value (EUR/MWh_th)	change	rationale
gas	24.57	16.25	−34%	Senegal-specific domestic gas price (\$5–6/MMBtu midpoint), replacing an international-hub or 2010-vintage DIW proxy
coal	9.55	14.90	+56%	Current API2 Rotterdam benchmark, replacing a 2010-vintage DIW figure roughly half today's price
biomass	7.41	14.40	+94%	Low end of EU agricultural-residue benchmark (Lamers et al. 2015), replacing a decade-old IEA 2011b figure
oil	52.91	<i>(unchanged)</i>	0%	Already sits between the global (Brent/HSFO) and Senegal-specific (government strategy note) benchmarks — no correction needed

Applied as a flat `costs.fuel:` override (not year-varying) across all twelve `config.uc_bess_{2025,2030,2035,2040,2045,2050}.yaml` / `config.uc_no_bess_{...}.yaml` files.

Coal and biomass were never built in any solved network in this project to date — their correction is a documentation/correctness fix, not expected to move any dispatch result. **Gas is the one correction with real leverage:** it directly sets CCGT/OCGT marginal cost, which sets how much thermal generation competes with BESS in every scenario.

8. Marginal cost — the downstream effect of the fuel correction

$\text{marginal_cost (EUR/MWh}_e) = \text{VOM (EUR/MWh}_e) + \text{fuel (EUR/MWh}_{th}) / \text{efficiency}$. VOM and efficiency were not changed (Sections 4 and 6), so the entire shift below comes from Section 7 alone. Figures use each carrier’s model efficiency and VOM as a representative constant (Sections 4/6); efficiency has a mild secondary year-dependence in the underlying DEA data that is unaffected by this correction and not re-derived here.

carrier	VOM (EUR/MWh)	efficiency	old marginal cost (EUR/MWh)	new marginal cost (EUR/MWh)	change
CCGT	4.23	60.0%	45.18	31.32	–31%
OCGT	4.76	43.0%	61.90	42.55	–31%
coal	3.26	35.6%	30.11	45.11	+50%
biomass	0.00	46.8%	15.83	30.77	+94%

Gas-fired generation (CCGT/OCGT) got meaningfully cheaper to dispatch — roughly 31% cheaper per MWh — which tends to *increase* how competitive thermal generation is against BESS in the model’s dispatch decisions, the opposite direction from what a naive “corrected figures always favour renewables” assumption might suggest. This is exactly the sensitivity the companion fuel-cost notebook flagged as “the single most consequential open question” — it has now been resolved with a Senegal-specific source rather than left as a caveat, and the direction of the effect (cheaper gas, not more expensive) is worth keeping in mind when interpreting the transmission/BESS comparison results run on top of this correction. Coal and biomass got correspondingly more expensive to dispatch, but since neither carrier is ever built in a solved network, this has no practical effect on current results.

9. BESS (storage) — summary only

Battery storage CapEx, FOM, VOM/degradation, and efficiency were recalibrated separately against NREL (Cole et al. 2025, *Cost Projections for Utility-Scale Battery Storage: 2025 Update*) and Ember (“How cheap is battery storage?”, Dec 2025), alongside two real software bugs that were found and fixed in the process (existing-battery efficiency defaulting to a lossless 100%, and new-build battery round-trip efficiency double-compounding). See [BESS_cost_assumptions.pdf/docx](#) for the full methodology, values, and sources — not repeated here.

10. Summary — what changed vs. what was checked

parameter	checked against	changed?
Investment (CapEx)	IRENA, Lazard, EIA AEO2022, UTS Africa Power Report, Taiba N'Diaye	No — onshore wind flagged as a sensitivity-run candidate

parameter	checked against	changed?
Lifetime	Indonesia reference case (MEMR 2024 / Athanasiou)	No
Efficiency	Indonesia reference case, EIA AEO2022 heat rates	No — biomass flagged as an anomaly worth checking upstream
Fixed O&M	Indonesia reference case, EIA AEO2022	No — onshore wind flagged, pairs with CapEx finding
Variable O&M	Indonesia reference case, EIA AEO2022	No — biomass zero-VOM flagged
Fuel cost	Senegal Gas-to-Power strategy note, API2 Rotterdam, Lamers et al. 2015, Brent/HSFO	Yes — gas, coal, biomass corrected; oil confirmed unchanged
Marginal cost	(derived)	Shifted as a direct consequence of the fuel correction — see Section 8

11. Sources

- **technology-data** project (v0.13.2) and the underlying Danish Energy Agency *Technology Data Catalogue for Generation of Electricity and District Heating*.
- DIW Berlin (2013), *Current and prospective costs of electricity generation until 2050*. <http://hdl.handle.net/10419/80348>
- IRENA (2024), *Renewable Power Generation Costs in 2024*.
- Lazard (2025), *Levelized Cost of Energy+ (LCOE+)*, v18.0.
- US EIA, *Annual Energy Outlook 2022*, Table 8.2, *Cost and Performance Characteristics of New Generating Technologies* (Sargent & Lundy basis).
- UTS (2025), *Africa Power Report: Senegal*.
- de Weijer, T. (TU Delft thesis), citing Indonesia's Ministry of Energy and Mineral Resources (MEMR) 2024 data and Langer et al. (2024).
- Yoga, R. (TU Delft thesis), citing the Indonesia Power Sector Technology Data Catalogue (Athanasiou catalogue).
- Republic of Senegal, Ministry of Petroleum and Energy, *Gas to Power* strategy note.
- API2 Rotterdam coal price index (Aug 2026 spot reference).
- Lamers, P. et al. (2015) — agricultural- and forest-residue feedstock cost ranges, as cited in UTS *Africa Power Report: Senegal*.
- IEA (2011b) — legacy biomass fuel-cost source, superseded by the above for this correction.
- Brent crude and Rotterdam/Singapore/Fujairah HSFO 380 bunker fuel spot references (Aug 2026).

Document prepared from the working session benchmarking generation technology cost inputs in the Senegal PyPSA-Earth model. All external figures were independently fetched/verified during that

session; see `BESS_Input_Cost_Benchmarking_2.ipynb` (Sections 1–3, 5–12) for the underlying computation and plots this document draws on.